7. Retail Billing with Multiple Discount Rules

Objective: To develop a Python program that processes multiple products, calculates quantities, applies category-wise taxes and discounts, and generates an itemized customer invoice.

1. Computational Thinking

- Decomposition: Divide the billing process into product entry, price calculation, discount calculation, tax calculation, and invoice generation.
- Pattern Recognition: Identify common product categories, tax rates, and discount rules.
- Abstraction: Focus on product name, price, quantity, category, and discount.
- Algorithm Design: Arrange the billing steps in a proper sequence.

2. Variables

Variable	Purpose
name	Product name
price	Price per item
quantity	Number of items
category	Product category
subtotal	Total product amount
discount	Discount amount
tax	Tax amount
total	Final bill amount

3. Data Types

- String – Product name and category.
- Integer – Quantity.
- Float – Price, tax, and discount.
- List – Stores multiple products.
- Tuple – Stores product details.

4. Operators

- Arithmetic: +, -, *, /

- Comparison: >, <, >=
- Logical: and, or

5. Algorithm

1. Start.
2. Read the number of products.
3. Create an empty product list.
4. Enter product name, price, quantity, and category.
5. Calculate subtotal for each product.
6. Apply category-specific tax.
7. Calculate the total bill.
8. Check discount eligibility.
9. Apply threshold discounts.
10. Limit the discount to the maximum concession amount.
11. Calculate the final payable amount.
12. Display the itemized invoice.
13. Stop.

6. Python Program

products = []n = int(input("Enter number of products: "))for i in range(n): print("\nProduct", i + 1) name = input("Enter product name: ") price = float(input("Enter price: ")) qty = int(input("Enter quantity: ")) category = input("Enter category (food/clothes/electronics): ") subtotal = price * qty if category == "food": tax_rate = 5 elif category == "clothes": tax_rate = 12 else: tax_rate = 18 tax = subtotal * tax_rate / 100 products.append((name, price, qty, subtotal, tax)) print("Subtotal:", subtotal)# Calculate billsubtotal = sum(p[3] for p in products)tax = sum(p[4] for p in products)total = subtotal + tax# Discount rulesif total >= 5000: discount = total * 0.20elif total >= 3000: discount = total * 0.10elif total >= 1000: discount = total * 0.05else: discount = 0# Maximum concession limitdiscount = min(discount, 1000)final_amount = total - discount# Invoiceprint("\n------- RETAIL INVOICE -------")for p in products: print(p[0], p[1], p[2], p[3], p[4])print("Subtotal:", subtotal)print("Tax:", tax)print("Discount:", discount)print("Final Amount:", final_amount)print("----------------------------")

7. Pseudocode

START

```
INPUT n
CREATE empty product list

FOR i = 1 TO n
    INPUT name, price, quantity, category
    subtotal = price * quantity

    IF category = food
        tax_rate = 5
    ELSE IF category = clothes
        tax_rate = 12
    ELSE
        tax_rate = 18
    END IF

    tax = subtotal * tax_rate / 100
    STORE product details
END FOR

CALCULATE total subtotal
CALCULATE total tax
total = subtotal + tax

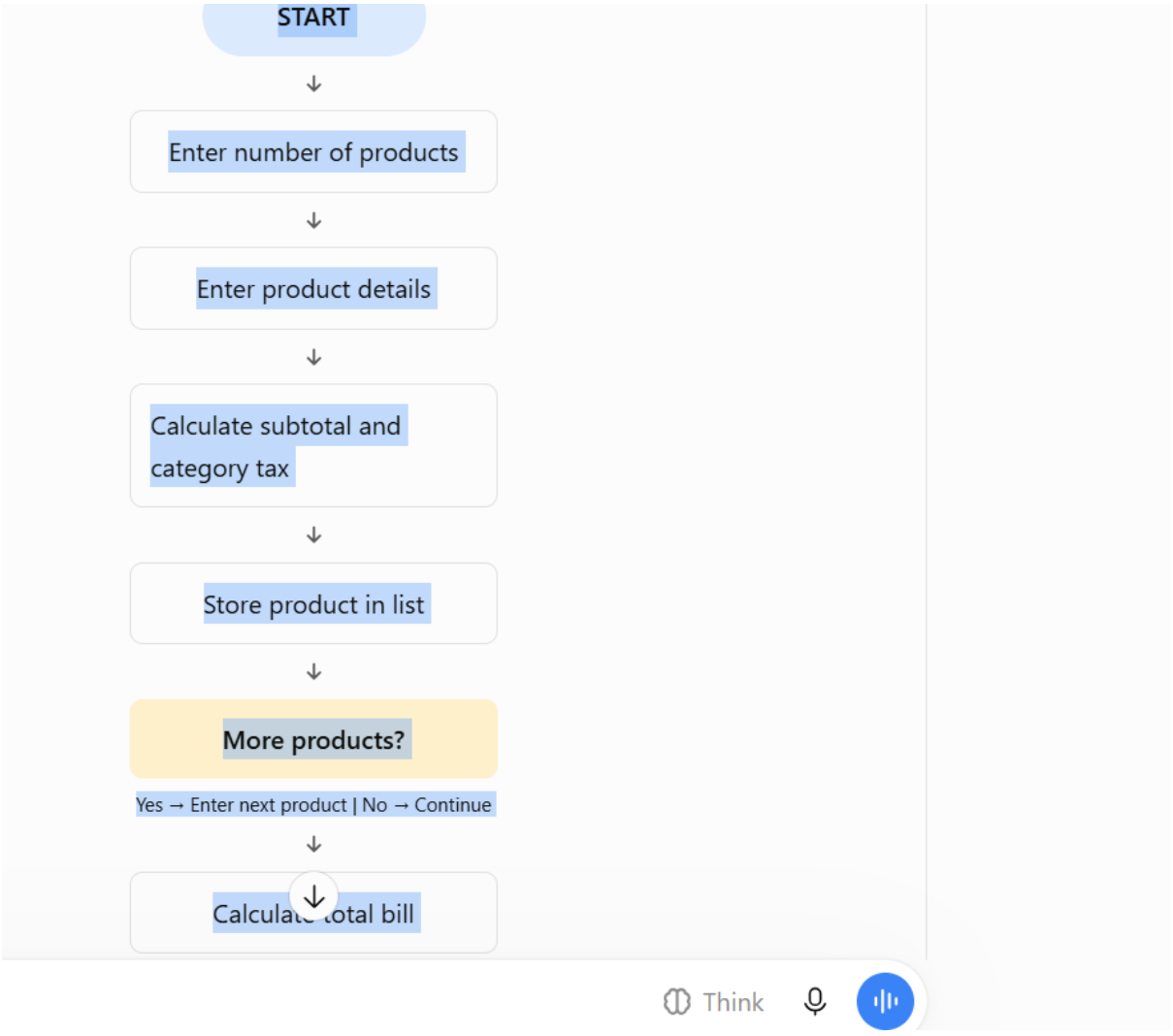
IF total >= 5000
    discount = total * 20 / 100
ELSE IF total >= 3000
    discount = total * 10 / 100
ELSE IF total >= 1000
```

```
    discount = total * 5 / 100
ELSE
    discount = 0
END IF

discount = MIN(discount, 1000)
final_amount = total - discount

DISPLAY ITEMIZED INVOICE

STOP
```



START

Enter number of products

Enter product details

Calculate subtotal and category tax

Store product in list

More products?

Yes → Enter next product | No → Continue

Calculate total bill

Apply discount rules

Apply maximum concession limit

Generate itemized invoice

STOP

9. Sample Output

Enter number of products: 2

Product 1

Enter product name: Rice

Enter price: 1000

Enter quantity: 2

Enter category: food

Product 2

Enter product name: Shirt

Enter price: 1500

Enter quantity: 1

Enter category: clothes

------- RETAIL INVOICE -------

Rice 1000 2 2000 100

Shirt 1500 1 1500 180

Subtotal: 3500

Tax: 280

Discount: 378

Final Amount: 3402

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10. Conclusion

This retail billing system uses Python concepts such as variables, data types, operators, conditional statements, loops, strings, lists, and tuples to calculate discounts, taxes, and generate an accurate customer invoice.